

COMPUTER SCIENCE PIONEERS



CSCI 1101 - Introduction to Computer
Science

Charles Babbage

Babbage, an English mathematician, was an English mathematician, inventor, and mechanical engineer, often referred to as the "**father of the computer**." He is best known for conceptualizing the first mechanical computer, which laid the groundwork for more complex machines in the future.

Babbage's most famous inventions are:

- The **Difference Engine**: Conceived in the 1820s
- The **Analytical Engine**: Designed in the 1830s

<https://www.britannica.com/biography/Charles-Babbage>



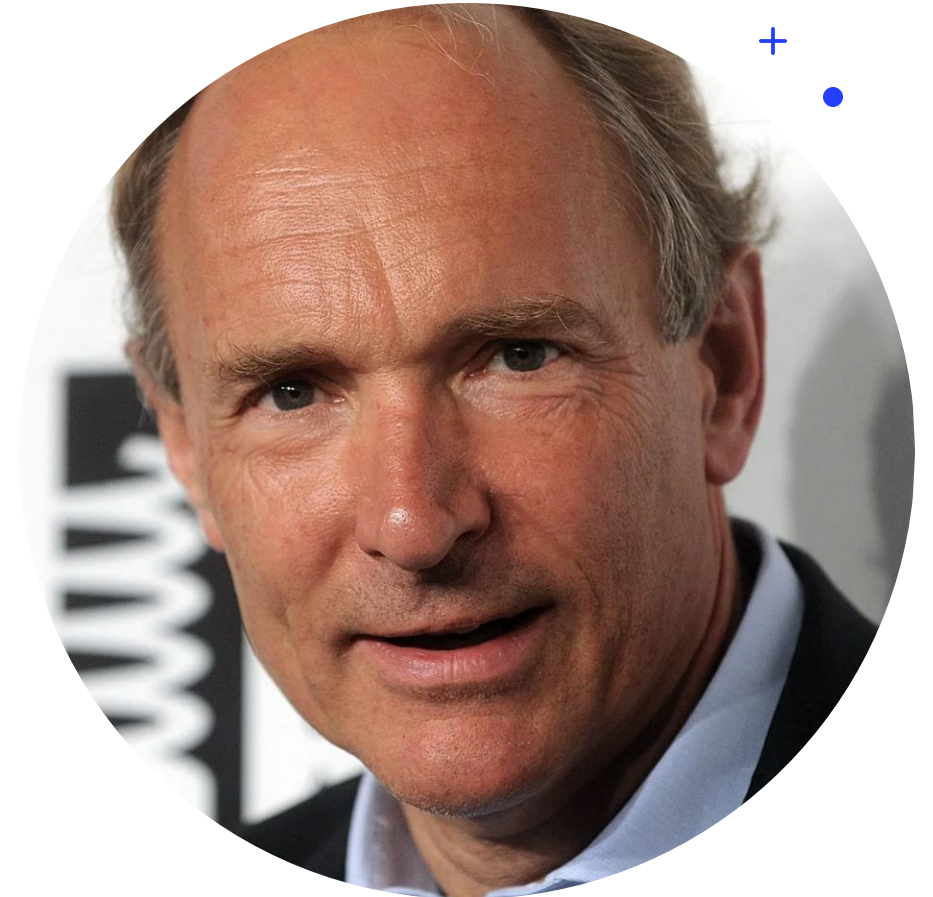
Tim Berners-Lee

Sir Tim Berners-Lee is a British computer scientist renowned for inventing the **World Wide Web**, revolutionizing communication and information dissemination.

His groundbreaking work, initiated in 1989, led to the development of the first web browser and server, laying the foundation for the interconnected digital landscape we navigate today.

Berners-Lee's contributions to computer science have had a profound and lasting impact, shaping the way people access and share information globally.

<https://www.britannica.com/biography/Tim-Berners-Lee>



George Boole

George Boole, a mathematician and logician, made significant contributions to computer science by developing **Boolean algebra**, a mathematical system that laid the foundation for digital logic and binary code.

His work became fundamental to the design of electronic circuits and the development of computer architecture.

<https://www.britannica.com/biography/George-Boole>



Grace Hopper

Grace Hopper, a pioneering computer scientist and naval officer, played a crucial role in the early development of computer programming languages.

She was instrumental in the creation of **COBOL** (Common Business-Oriented Language), a programming language still in use today for business, finance, and administrative systems.

<https://www.britannica.com/biography/Grace-Hopper>



Ada Lovelace

Ada Lovelace, often regarded as the **world's first computer programmer**, made pioneering contributions to computer science in the 19th century.

Recognized for her work on Charles Babbage's Analytical Engine, she wrote detailed notes that included the first algorithm intended for implementation by a machine.

<https://www.britannica.com/biography/Ada-Lovelace>



Bjarne Stroustrup

Bjarne Stroustrup, a Danish computer scientist, is widely known for creating the **C++** programming language.

His work on C++, an extension of the C programming language, has had a profound impact on software development, providing features like object-oriented programming and templates.

<https://engineering.tamu.edu/electrical/profiles/stroustrup-bjarne.html>



Linus Torvalds

Linus Torvalds, a Finnish-American software engineer, is best known for creating the **Linux** operating system kernel.

His development of Linux, an open-source Unix-like operating system kernel, has been instrumental in the growth of free and open-source software.

<https://www.britannica.com/biography/Linus-Torvalds>



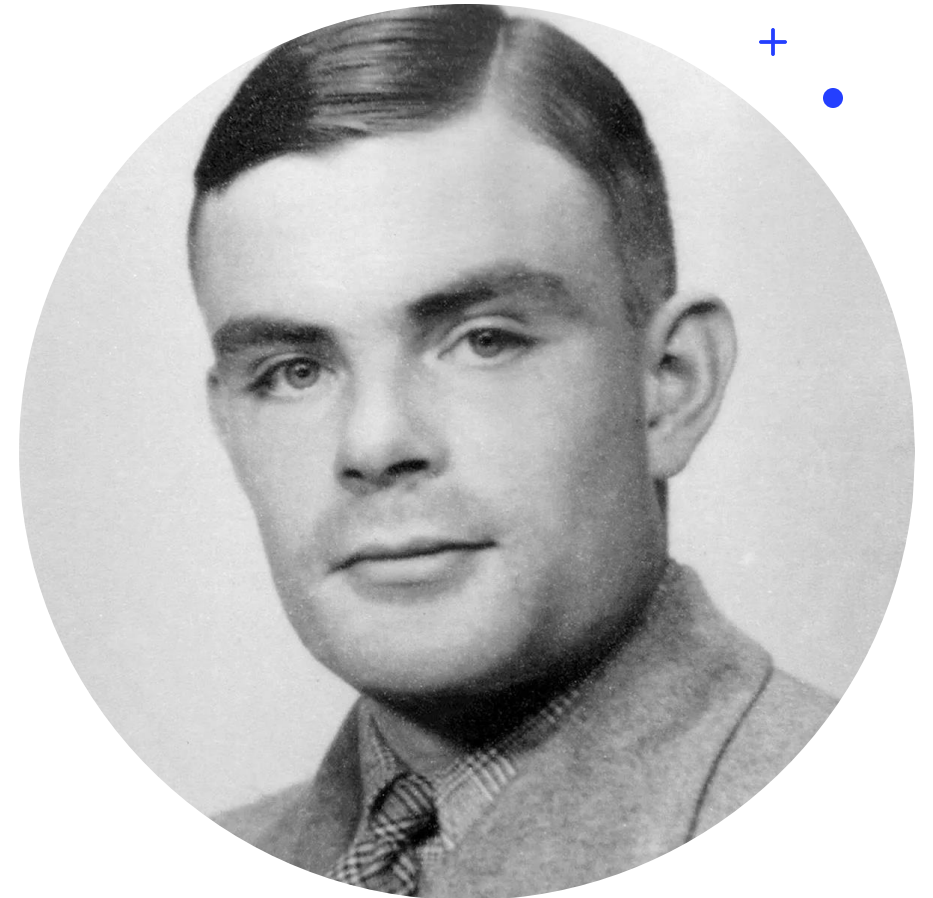
Alan Turing

Alan Turing, a pioneering English mathematician and computer scientist, is widely recognized as one of the **fathers of modern computing** and **artificial intelligence**.

During World War II, Turing's efforts cracked the German Enigma code

Turing developed the "Turing machine", a machine capable of processing a stream of 1s and 0s (binary) on tape to solve any problem. This theoretical concept laid the groundwork for modern computers.

<https://www.britannica.com/biography/Alan-Turing>



Bill Gates

Bill Gates while at Harvard University, he developed a programming language for his computer and decided to drop out of university to concentrate on this project

Gates started a company called **Microsoft** and made a fortune from **MS-DOS**.

Over the next few years Gates made billions of dollars and has donated a lot of his fortune to improving the lives of people in developing countries.



Steve Jobs

Steve Jobs also dropped out of university at the age of 21 to start his company **Apple**.

Apple I was one of the first home computers and was sold for \$600 and made a fortune with the invention of the Graphical User Interface (GUI) in **Apple II**.

Gates became a millionaire at the age of 25.



Steve Wozniak

Steve Wozniak, **co-founder of Apple** Inc., is a key figure in the history of personal computing.

Often referred to as "The Woz," he designed the Apple I and Apple II computers, which played a pivotal role in the microcomputer revolution.

<https://www.britannica.com/biography/Stephen-Gary-Wozniak>

